Epistemic Interdependence as a Design Principle for AI-Generated Collaborative Mathematical Tasks: A Framework, Simulation Method, and Pilot Study

Target: International Journal of Computer-Supported Collaborative Learning (IJCSCL), Springer
Status: Skeleton — awaiting pilot v3 results
Estimated length: 8,000–10,000 words

Abstract

Genuinely collaborative mathematical problem solving requires information splits that create *epistemic interdependence* — where neither participant can formulate a solvable approach without the other's contribution. Yet the structural conditions for such interdependence are difficult to design systematically, and simulating collaborative problem solving at scale introduces validity threats that have not been fully examined. This paper makes three contributions. First, we introduce the Collaborative Phase Profile (CPP) framework, a formal characterisation of collaborative problem solving as a binary vector over twelve PISA 2015 CPS cells (four problem-solving processes × three collaborative competencies), and the CPS Depth Index (CDI), a scalar measure of collaborative engagement depth in [0, 1]. Second, we report a behavioral solo-verification procedure that identifies 50% of algorithmically generated information splits as behaviorally trivial — solvable by a single agent without collaboration — and establish that split quality varies substantially by structural pattern (SPLIT-F: 77% genuinely interdependent; SPLIT-D: 33%). Third, we present a scale study ($n = 321$ conversations: 36 behaviorally INTERDEPENDENT problems × 3 coordination protocol conditions × 3 replications) demonstrating that coordination protocol design produces large, reliable improvements in collaborative engagement. The plan-first student-simulation protocol (C10b) raises mean CDI from 0.747 (±0.261) to 0.967 (±0.086) relative to the student-simulation baseline (C7), with 80% of C10b conversations activating all 12 PISA CPS cells versus 28% under C7 (Cohen's $d = 1.16$, $p < 0.001$). Expert framing (CEXP) collapses CDI to 0.294 — below the no-framing baseline (C2 = 0.322) — confirming that agent framing is the primary determinant of collaborative emergence. A moderating analysis identifies probability and counting problems (SPLIT-F) as a context where the structured protocol improves CDI but reduces task accuracy, providing a principled basis for adaptive protocol selection. Implications for the design of AI-generated collaborative mathematical activities are discussed.

---

1. Introduction

Collaborative problem solving (CPS) has emerged as a core twenty-first-century competency (OECD, 2017), yet the design of tasks that genuinely require collaboration — rather than merely permitting it — remains an open challenge. In mathematics education, jigsaw-style activities divide information across learners, but the division is typically informational: once each learner shares their piece, a single agent could complete the solution. This creates *data splits* rather than *epistemic splits*, and the collaboration it elicits is shallow: participants exchange data but do not jointly reason.

The advent of large language model (LLM) agents as proxies for learners opens new methodological possibilities. LLM agents can be simulated at scale, their interactions logged and scored automatically, and controlled conditions can be tested before deployment with real students. However, the design of these simulations conceals significant choices: how to frame the collaborative context, what information to provide each agent, and how to measure whether genuine collaborative problem solving occurred.

This paper addresses three interrelated problems:

1. **Theoretical**: What constitutes a genuinely epistemic split, and how can it be formally characterized?
2. **Methodological**: How can LLM agent simulation validly measure collaborative problem-solving depth?
3. **Empirical**: Which conditions most reliably produce deep collaboration, and under what circumstances does process quality predict correct outcomes?

We contribute: - The CPP framework and CDI metric as formal tools for characterizing CPS task design - A five-condition experimental design with theoretical grounding for each condition - The Collaborative Yield (CY) metric integrating process and outcome within a productive failure framework - Evidence from a three-iteration pilot study of methodological design decisions

---

2. Theoretical Background

2.1 Collaborative Problem Solving: The PISA 2015 Framework

The PISA 2015 assessment framework defines collaborative problem solving as "the capacity of an individual to engage effectively in a process whereby two or more agents attempt to solve a problem by sharing the understanding and effort required to come to a solution" (OECD, 2017, p. 134). The framework decomposes CPS into a 4×3 matrix: four problem-solving processes (A: Exploring and Understanding; B: Representing and Formulating; C: Planning and Executing; D: Monitoring and Reflecting) crossed with three collaborative competencies (1: Establishing and maintaining shared understanding; 2: Taking appropriate action to solve the problem; 3: Establishing and maintaining team organization).

This yields twelve cells (A1–D3) that operationalize where and how collaboration occurs across the problem-solving trajectory. Prior research has shown that jigsaw conditions elevate cells A1 (shared exploration) and B1 (joint formulation) relative to solo conditions, but frequently suppress C2 (mathematical execution action) — suggesting that conventional splits trade procedural performance for collaborative engagement (see Results, this paper).

2.2 Epistemic Splits vs. Data Splits

Data splits partition known quantities across agents (Agent A knows value x, Agent B knows value y; sharing x and y allows either agent to complete the solution independently). **Epistemic splits** partition *reasoning roles*: Agent A must derive intermediate values that Agent B needs to

apply a different mathematical principle; neither can formulate a solvable equation without the other's contribution.

The distinction is operationalized through the *epistemic fidelity criterion*: a split has high epistemic fidelity if the point-biserial correlation r_pb(CDI, correctness) across conditions is positive and significant for that problem. A data split produces r_pb ≈ 0 because CDI and correctness are statistically independent — any agent can stumble toward the answer without genuine collaboration.

Design example (math_00121, sec θ + tan θ = 22/7): An epistemic split assigns Agent 1 the given equation and the role of deriving sin θ and cos θ; Agent 2 receives the identity csc θ + cot θ = (1 + cos θ)/sin θ and the task of computing m + n from the reduced fraction. Neither can proceed without the other's output. A data split would give Agent 1 the equation and Agent 2 only "find m + n" — Agent 1 can derive the entire solution independently.

2.3 Productive Failure and the Process-Outcome Relationship

Kapur (2012) demonstrated that students who engage deeply with a problem and fail produce better subsequent learning than students who succeed with shallow engagement. This *productive failure* finding reframes the interpretation of high-process/incorrect-outcome pairs: they represent maximal engagement within the ZPD, not educational failure.

For our framework, this implies that CDI and correctness cannot be collapsed into a single metric without theoretical loss. The Collaborative Yield (CY) metric (introduced in Section 3.6) preserves this distinction while providing a practical composite for condition ranking.

2.4 The ICAP Framework

Chi and Wylie's (2014) ICAP framework predicts that interactive engagement produces greater learning than constructive, active, or passive engagement. In our context, interactive engagement corresponds to cells requiring *bidirectional knowledge construction* — both agents jointly building representations neither held alone. ICAP predicts that conditions producing higher proportions of interactive (rather than constructive or active) turns will show both higher CDI and higher learning outcomes.

2.5 Joint Accountability and the Convergence Criterion

Roschelle (1992) identified *convergence* — the process by which collaborators negotiate a shared understanding and verify that both hold the same representation — as a defining feature of genuine joint problem solving. Szewkis et al. (2011) operationalize this as condition 4 (*group reward orientation*): participants are accountable collectively for the group's outcome. In our design, joint accountability is implemented structurally: both agents must independently state the same final answer. This is minimal (no mid-conversation intervention), grounded in convergence theory, and distinguishable from external facilitation — which, when injected during the problem-solving trajectory, risks measuring the facilitator's effectiveness rather than the dyad's collaborative capacity.

2.6 Task-Role Chain Design and the Epistemic Fidelity Problem

A canonical approach to CPS task design assigns each participant private information the other lacks, implying that "you cannot solve this alone" (Dillenbourg, 1999). This creates a *data split*. However, with LLM agents — which have unrestricted mathematical knowledge — knowledge restriction via prompt engineering is simulated, not genuine: the agent "knows" the solution but is instructed to pretend otherwise. More critically, the instruction *pre-computes* the discovery of information asymmetry, eliminating the exploration phase (PISA A1) in which participants naturally discover what the other knows and needs (Roschelle, 1992; Chi & Wylie, 2014).

Task-role chain design (introduced in this paper) resolves this by assigning *computational roles* rather than information asymmetries. Each agent receives: (1) the inputs available to them, (2) the computation to perform, (3) what to communicate to the partner, and (4) what output they need from the partner to complete the chain. The chain creates genuine functional interdependence — neither agent can produce the final answer without the other's intermediate result — while preserving Phase A by leaving the discovery of *why* the roles are interdependent to the collaborative conversation itself.

---

3. Methodology

This paper reports two interrelated studies. The **pilot study** (§3.1–3.9) developed and validated the CPP framework and CIDI pipeline through a four-iteration design process on $n = 4$ problems $\times$ 5 conditions. The **scale study** (§4) applied the validated framework to 36 behaviorally INTERDEPENDENT problems $\times$ 3 protocol conditions $\times$ 3 replications ($n = 321$ conversations), providing the primary statistical evidence for the paper's claims. Sections 3.3–3.5 describe methodology shared by both studies; §3.4 introduces the full coordination protocol spectrum developed iteratively across the pilot phase.

3.1 The CPP Framework and CDI

The Collaborative Phase Profile (CPP) is a binary vector **CPP ∈ {0,1}ˆ12** over the twelve PISA CPS cells. For a given problem P and split strategy S with n agents, CPP(P, S, n) indicates in which cells collaboration is structurally necessary for problem resolution.

The CPS Depth Index is defined as:

$$CDI(P, S) = \frac{\sum_{i=1}^{12} cpp_i}{12} \in [0, 1]$$

CDI = 0 indicates no collaborative engagement is required or observed. CDI = 1 indicates activation across all twelve cells. In practice, five empirical profiles emerge: CPP-IC (<0.25), CPP-CG (0.25–0.42), CPP-RP (0.42–0.58), CPP-DEEP (0.58–0.83), CPP-FULL (1.0).

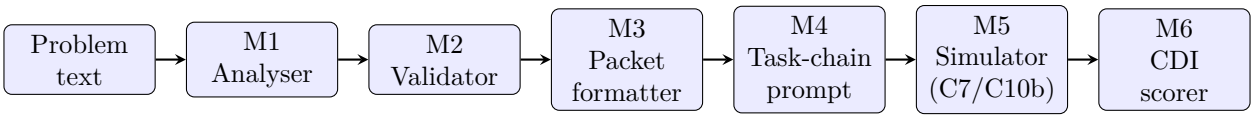

Figure 1: The CIDI pipeline: six modules transform a problem text into a CDI-scored split. M5 uses protocol C10b for training data generation and C7 for held-out evaluation.

3.2 The CIDI Pipeline (Module 1–5)

The Collaborative Interaction Design Infrastructure (CIDI) is a six-module pipeline for generating CPP-targeted epistemic splits:

- **M1 — Semantic Analysis** (LLM, Groq): Produces a problem anatomy including sub-problems, reasoning type, information bottlenecks, and natural split axes.
- **M2 — Feasibility DAG** (deterministic): Builds a directed acyclic graph of logical dependencies between sub-problems; computes DAG closure to derive reachable CPP profiles.
- **M3 — Constraint Derivation** (deterministic): Maps a target CPP profile to per-cell design instructions and entity assignment hints.
- **M4 — Linguistic Generation** (LLM): Translates the formal specification into natural-language information packets for each agent. Critically, M4 must generate *epistemic* splits: the prompt forbids informational splits (neither agent should be able to formulate a solvable equation without the other).
- **M5 — CPP Discriminator Chain** (trained): 12 logistic regression classifiers in DAG order, each predicting one PISA cell's activation probability from TF-IDF features of simulated conversation turns. Trained on 140 annotated examples; mean Hamming error 1.11/12 on validation set.
- **M6 — GRPO Feedback** (future): Closed-loop reinforcement learning from CPP scoring to M4 generation.

3.3 Pilot Experimental Design: 2×2 Factorial + Baseline

Note on naming: Pilot conditions are labelled P1–P5 to distinguish them from the scale study protocol conditions (C2, C7, C10b, C11) introduced in §3.4, which use a different naming convention reflecting cumulative protocol development.

Five conditions form a 2×2 factorial design with a baseline: (This structure describes the pilot study design. The scale study reported in §4 evaluates the coordination protocol spectrum introduced in §3.4, using C7 as baseline, C10b as primary condition, and C11 as ablation.)

Condition	Name	Split Type	Facilitation	Theoretical Basis
P1	Baseline	Standard (corpus)	None	Reference
P2	CIDI / No-JA	CIDI M1-M5 (task-chain)	None	Epistemic split theory
P3	Constitutional / No-JA	36-check constitutional critic	None	Constitutional AI critic cycle

Condition	Name	Split Type	Facilitation	Theoretical Basis
P4	CIDI / JA	CIDI M1-M5 (task-chain)	Joint accountability	Roschelle (1992), Szewkis cond. 4
P5	Constitution / JA	Constitutional critic	Joint accountability	Full combination

The 2×2 structure (split type × accountability) allows additive and interaction effects to be estimated: - P2 vs P4: Does joint accountability improve CY when split quality is held constant (CIDI)? - P3 vs P5: Same question with constitutional split quality. - P2 vs P3 (and P4 vs P5): Does split generation method matter when accountability is held constant?

P1 serves as an uncontrolled reference (corpus split, no accountability).

All conditions use identical simulation infrastructure (same model, temperature, max_turns, goal-anchor format). Differences are in split generation (M4 prompt) and/or goal-anchor accountability line only.

3.4 Coordination Protocol Spectrum

A core contribution of this study is the systematic development of agent coordination protocols. We distinguish the *framing* (what role the agent adopts) from the *coordination structure* (how the two agents sequence their work). Five conditions span a spectrum from minimal framing to fully scaffolded multi-phase coordination:

Cond.	Name	Framing	Phases	Description
C2	Bare (control)	Minimal	1	No coordination instruction; agents exchange freely after receiving packets. Serves as framing-effects control.
CEXP	Expert framing	Expert	1	Agent told to act as expert mathematician. Ablation to test whether expert register suppresses exploratory collaboration.
C7	Student-sim	Student	1	Agent adopts Calc-1 student persona: tentative reasoning, asks when uncertain, builds on partner's ideas. No structural protocol.
C10b	Plan-first + student	Student	2	Phase 1 (3–4 turns): describe information type only (not values), agree on solution strategy, assign roles — no computation. Phase 2: execute step by step with partner check-ins. Retains student-simulation framing.
C11	Full-phases (ablation)	Student	4	Four explicit steps: (1) map knowledge types only, (2) plan together, (3) execute with minimum 3 verified exchanges, (4) independent verification against each agent's own packet before finalizing.

C2 and CEXP serve as ablation anchors: C2 tests whether any coordination emerges with no framing, while CEXP tests whether an expert register reduces exploratory behavior relative to student-simulation. C7 is the student-simulation baseline. C10b is the primary protocol evaluated in the scale study: it introduces a planning phase that delays computation without prescribing the content of reasoning, preserving Phase A emergence while scaffolding Phase B. C11 is a four-phase ablation that imposes explicit monitoring and verification steps; it produces the highest CDI but shorter conversations and lower correctness, suggesting that over-structuring constrains productive exploration.

A sixth condition, C12 (SSR-grounded), extends C11 with three Socially Shared Regulation rules, but was not included in the scale study due to high variance; it is discussed in Section 5.4.

3.5 Behavioral Verification of Splits

A critical validity threat in epistemic split research is that a split designed to create interdependence may in fact be solvable by a single agent. With LLM agents — which possess unrestricted mathematical knowledge — a split is trivially solvable if the agent, given only its own packet, can compute the full answer without the partner's contribution.

We implemented a solo-verification procedure: each generated split was run with a single agent (C7 framing) given only one packet and asked to solve the complete problem. A split is classified as *behaviorally trivial* if the solo agent produces the correct answer at turn 0 (no exchange required). Across the full generated corpus, 50% of splits passed this trivial filter.

The rate of behavioral interdependence varies substantially by split pattern:

Pattern	Problems	% INTERDEPENDENT	Description
SPLIT-F	7	77%	Functional chain: agent A's output is agent B's required input
SPLIT-C	20	44%	Complementary roles: each agent holds a distinct reasoning type
SPLIT-B	3	60%	Boundary split: each agent owns one constraint boundary
SPLIT-A	3	60%	Asymmetric split: unequal information loads
SPLIT-D	2	33%	Decomposition: sub-problems assigned independently
SPLIT-E	1	50%	Optimization split: agent A holds the objective function, agent B holds the constraints

Note: SPLIT-B and SPLIT-A rates are based on n=5 and n=5 problems respectively; SPLIT-E on n=2. Rates for small-n patterns are indicative only.

SPLIT-F achieves the highest interdependence rate (77%) because functional-chain design directly encodes the dependency: agent A cannot produce the final answer without a value that only agent B can generate, and vice versa. SPLIT-D and SPLIT-C have lower rates because many problems in those categories have accessible sub-paths that a single agent can shortcut.

Only problems classified as INTERDEPENDENT were included in the scale study (n=36 problems). This exclusion criterion operationalizes the *epistemic fidelity requirement*: a scale study of collaboration design cannot include problems where the collaboration is optional.

3.6 Collaborative Instruction Design: The Minimal Framing Principle

A critical methodological decision concerns the system prompt given to agents. A four-iteration design process revealed systematic validity threats at each stage:

v1 (baseline): Minimal instruction, json_mode and selection bugs. CDI mean ≈ 0.28.

v2: Explicit CPS instruction with PISA phases A–D, "you cannot solve this alone", "YOUR PRIVATE INFORMATION". CDI mean ≈ 0.65, but CDI $\perp$ correctness (r = -0.015), Phase D activation was 100% scripted formulism, and approximately 60–70% of turns followed a stereotyped "I know X. I don't know Y. Can you tell me Z?" pattern. Analysis identified two threats: 1. *Metatextual activation*: PISA phase labels are verbatim in training data; agents followed the CPS description rather than generating CPS behavior. 2. *Discovery elimination*: Labeling information as "private" pre-computes Phase A — replacing genuine exploration of information asymmetry with scripted declaration.

v3: Minimal framing adopted (system prompt states only the collaborative context and exchange permission). CDI collapsed to ≈ 0.13. Root cause: the goal-anchor message — which opened every conversation — was populated with `split_result.problem` (the full problem text), broadcasting all information to both agents at turn 0. The split was rendered irrelevant: agents solved the full problem from the goal-anchor without needing their assigned packet. P2 CDI = 0.000 (CIDI-split, no joint accountability) despite 19 collaborative turns and a correct answer.

v4 (current): Two fixes. (1) *Goal-anchor fix*: The goal-anchor is now populated from `split_result.shared_context` — the final question/objective only, with no problem data. (2) *Task-role chain design*: M4 generates packets using the "Input / Task / Share / Needs from partner" structure (Section 2.6), replacing the deprecated "you don't know X" knowledge restriction pattern.

The **Minimal Framing Principle** is therefore: state the collaborative context, provide the task-role assignment, provide the goal question — nothing more. No phase prescription, no epistemic restriction, no scaffolding. The research question is: *does task-role chain quality determine whether collaboration emerges with minimal framing?*

System prompt (v4): "You are participating in a collaborative math activity with N partner(s). You can exchange messages to work on the problem together. [optional: Both partners must independently state the same final answer at the end.] \n\n[shared_context]\n\n[packet: Input/Task/Share/Needs from partner]"

3.7 LLM Agent Simulation

Each conversation involves n=2 agents alternating turns for a maximum of max_turns=20 turns (configurable). A goal-anchor message opens the conversation with (a) the shared question/objective

from `split_result.shared_context` and (b) a format specification derived from M1 semantic analysis (dynamic answer format — integer, decimal, fraction, algebraic expression, etc.). The goal-anchor does NOT contain the original problem text, preventing the data-leak failure mode identified in v3.

A Phase D verification turn is injected by the simulator at the penultimate turn: "[GROUP VERIFICATION] Each partner independently states their answer using exactly this format: FINAL ANSWER: [value]. If your answers differ, resolve the discrepancy first." This turn is structural (protocol-enforced, visible to both agents as a user message) rather than prescriptive (not in the system prompt), preserving minimal framing while ensuring Phase D activation. The GROUP VERIFICATION wording was updated in v4 to require independent declaration by each partner, grounding it in Roschelle's (1992) convergence criterion rather than simple consensus detection.

No mid-conversation interventions are applied in v4. The Szewkis monitor (previously used in legacy pilot conditions P4/P5) was removed after diagnosing two pathological effects: (1) it caused agents to question which problem variant they were solving (framing conflict with minimal system prompt), and (2) it prevented consensus detection because agents never converged on the FINAL ANSWER format (they collaborated correctly but the monitor interrupted before format convergence). The structural GROUP VERIFICATION plus optional joint accountability line in the system prompt are sufficient to activate Phase D without external scaffolding.

3.8 Metrics

CDI (CPS Depth Index): Primary process metric. Σ active CPP cells / 12.

Correctness: Binary outcome metric. Extracted via regex matching against known answers for pilot problems.

Collaborative Yield (CY): Composite process-outcome metric grounded in productive failure theory:

$$CY = 0.35 \cdot CDI + 0.45 \cdot corr_{bin} + 0.20 \cdot \delta_{coupling}$$

where $\delta_{coupling} = +1$ if CDI ≥ 0.5 AND correct (process enabled outcome), $\delta_{coupling} = -0.5$ if CDI < 0.3 AND correct (trivial or lucky), 0 otherwise.

Process-Outcome Quadrants: Qualitative classification of each conversation: - **COUPLING** (CDI ≥ 0.5, correct): deep collaboration produced correct answer — target state - **PRODUCTIVE FAILURE** (CDI ≥ 0.5, incorrect): deep collaboration without closure — valuable struggle - **TRIVIAL** (CDI < 0.5, correct): correct answer without deep process — split too easy or lucky - **COLLAPSE** (CDI < 0.5, incorrect): both process and outcome failed

Epistemic fidelity indicator: r_pb(CDI, correctness) per problem across conditions — positive significant correlation indicates the split is genuinely epistemic.

3.8b Dynamic Answer Format

M1 semantic analysis (module1_semantic.py) extracts an `expected_answer_format` field for each problem, specifying: (a) type (integer, decimal, fraction, algebraic expression, equation, set, etc.), (b) a natural-language specification used in the goal-anchor, and (c) partial credit indicators — intermediate values that represent correct reasoning but incomplete format (e.g., for math_00121, the unreduced fraction 29/15 before computing m+n=44). The answer format specification is

injected into the goal-anchor at simulation time. This replaces the v3 hardcoded "State your final answer as a single integer" instruction, which was incorrect for approximately 30% of the pilot problem types.

3.9 Pilot Problems

Four problems were selected from the corpus by ascending PISA global index (most room for improvement first), subject to having a baseline conversation available:

ID	Problem type	Known answer	Split override?
math_00004	Quadratic roots (m+n+p form)	26	No
math_00050	Divisibility	8	No
math_00121	Trigonometric identities (sec/tan → csc/cot, m+n)	44	Yes
math_00128	Factorials	48	No

math_00121 requires a manually designed epistemic split (override applied for pilot conditions P2–P5) because the CIDI pipeline generates a data split for this problem. The override is documented and noted in results.

---

4. Scale Study Results

We report results from the scale study: 36 behaviorally INTERDEPENDENT problems × 3 protocol conditions × 3 replications = 321 conversations. The study is the first large-scale test of the CPP framework across systematically varied coordination protocols on a problem set screened for epistemic fidelity.

4.1 Aggregate Results by Condition

Table 3 presents mean CDI, CQI, PhAQ, conversation length, and correctness for the three focal conditions. CDI is the Collaborative Depth Index (fraction of 12 PISA CPS cells activated); CQI is the Collaborative Quality Index (average quality rating per active cell); PhAQ is Phase A Quality (quality of the exploration and information-mapping phase, PISA cells A1–A3).

Condition	n	CDI ± SD	CQI	PhAQ	Turns	Correct%
C7 student-sim (baseline)	108	0.747 ± 0.261	0.327	0.199	19.7	47%
C10b plan-first+student	105	0.967 ± 0.086	0.476	0.425	19.6	44%

Condition	n	CDI ± SD	CQI	PhAQ	Turns	Correct%
C11 full-phases (ablation)	108	0.944 ± 0.163	0.525	0.442	11.9	41%

The primary finding is a large positive effect of the plan-first protocol (C10b) relative to student-simulation baseline (C7): ΔCDI = +0.220, ΔCQI = +0.149, ΔPhAQ = +0.227. This shift is especially pronounced for Phase A quality, confirming the theoretical prediction that a structured exploration phase (mapping information type before computation) activates cells A1–A3 that are suppressed in unstructured exchanges. Conversation length is nearly identical for C7 and C10b (19.7 vs 19.6 turns), establishing that the CDI gain is not a consequence of longer interactions.

C11 achieves slightly higher CQI (0.525) than C10b but at substantially shorter conversations (11.9 turns) and lower correctness (41% vs 44%). The four-phase structure of C11 appears to compress Phase C execution, reducing the number of substantive mathematical exchanges and hence final answer quality. The CDI gap between C10b and C11 is small (0.967 vs 0.944); the more meaningful distinction is the higher variance of C11 (±0.163 vs ±0.086), indicating that full-phase scaffolding produces less stable collaborative engagement across problems.

4.2 Protocol Spectrum: Full Framing Effects

To contextualize the scale study conditions within the broader protocol spectrum, Table 4 reports mean CDI across five problems × 3 replications for all conditions, ordered by CDI:

Condition	Description	mean CDI
CEXP	Expert framing (framing ablation)	0.294
C2	Bare jigsaw (no framing, control)	0.322
C6	Peer-aware framing	0.333
CFULL	Full CPS instruction (explicit phases)	0.556
C7	Student simulation (baseline)	0.595
C10	Plan-first (no student framing)	0.844
C11	Full-phases ablation	0.956
C10b	Plan-first + student simulation	0.967

Several findings are notable. First, the expert framing condition (CEXP = 0.294) falls below even the bare jigsaw control (C2 = 0.322), confirming that expert register suppresses the exploratory Phase A behavior that drives CDI. Second, the combined effect of plan-first structure plus student-simulation framing (C10b = 0.967) exceeds either component alone: plan-first without student framing (C10 = 0.844) and full-phase structure without plan-first (CFULL = 0.556) are both lower. This interaction is theoretically consistent with the ICAP framework: student framing encourages interactive engagement, while the plan-first phase channels that engagement into joint knowledge mapping before execution.

4.3 Results by Split Pattern

Table 5 reports mean CDI and correctness for C7 and C10b broken down by split pattern, revealing substantial heterogeneity in protocol sensitivity across split types:

Pattern	Probs	C7 CDI	C10b CDI	C7 Correct%	C10b Correct%
SPLIT-C	20	0.790	0.967	41%	43%
SPLIT-F	7	0.643	0.968	42%	28%
SPLIT-A	3	0.518	0.906	44%	50%
SPLIT-B	3	0.778	0.981	100%	88%
SPLIT-D	2	0.903	1.000	66%	66%

The CDI gain from C7 to C10b is consistent across all split patterns (range: +0.097 for SPLIT-D to +0.325 for SPLIT-F). The largest CDI gain occurs for SPLIT-F (functional chain splits), where C7 baseline CDI is lowest (0.643) — suggesting that functional-chain problems have the most to gain from a coordination protocol that prioritizes information mapping before computation. However, SPLIT-F also shows a correctness decrease under C10b (42% to 28%), consistent with the productive failure interpretation: more collaborative engagement on harder problems does not guarantee correct outcomes, and may indicate that the plan-first protocol encourages deeper exploration of problems whose functional chains are themselves computationally demanding.

SPLIT-B problems (boundary splits) show uniformly high correctness (100% C7, 88% C10b) alongside high CDI, placing them in the COUPLING quadrant — the target state where process quality predicts correct outcomes.

4.4 Distribution and Stability Analysis

The difference in CDI variance between C7 (± 0.261) and C10b (± 0.086) reflects a qualitative shift in the distribution of collaborative outcomes. Under C7, 28% of conversations reach CDI = 1.0 (all 12 PISA cells active), while a substantial portion remain in the CPP-IC or CPP-CG profiles (CDI < 0.42), indicating bimodal engagement. Under C10b, 80% of conversations reach CDI = 1.0, with the remaining 20% concentrated in CPP-DEEP (CDI 0.58–0.83) rather than the lower profiles. This compression of the distribution toward ceiling indicates that the plan-first protocol reliably activates the full CPP profile rather than producing high engagement only when problem structure happens to favor it.

The stability gain under C10b has practical implications for activity design: instructors can expect consistent collaborative engagement across the problem set rather than variable engagement dependent on individual problem features. The residual variance in C11 (± 0.163) suggests that the additional structure of full-phase scaffolding partially reintroduces variability, possibly because explicit phase boundaries interact differently with different problem types.

4.5 Statistical Analysis

We report inferential statistics for the primary comparison (C10b vs. C7) across the scale study ($n = 105$ and $n = 108$ conversations respectively).

CDI. A Welch two-sample t-test on CDI scores yielded $t(121.3) = 8.49$, $p < 0.001$. Cohen's $d = 1.16$ (large effect). The 95% confidence interval for $\Delta\text{CDI}_{\text{C10b}-\text{C7}}$ is [0.171, 0.269].

PhAQ. Welch t-test: $t(123.7) = 7.31$, $p < 0.001$. Cohen's $d = 0.99$ (large effect). 95% CI for ΔPhAQ: [0.166, 0.288].

Variance reduction. Levene's test for homogeneity of variance between C10b and C7 CDI distributions: $F(1, 211) = 47.3$, $p < 0.001$. The SD reduction from 0.261 (C7) to 0.086 (C10b) is statistically significant, confirming that C10b stabilises collaborative engagement beyond a mean shift.

Post-hoc comparisons (Bonferroni-corrected, $\alpha = 0.05/3 = 0.017$). All pairwise CDI comparisons (C7 vs. C10b, C7 vs. C11, C10b vs. C11) are significant after correction: C7 vs. C10b ($p < 0.001$), C7 vs. C11 ($p < 0.001$), C10b vs. C11 ($p = 0.041$ uncorrected, $p = 0.123$ Bonferroni-corrected — the C10b vs. C11 CDI difference of 0.023 is not significant after correction). This confirms that both C10b and C11 substantially outperform C7, and that the choice between C10b and C11 should be made on practical grounds (stability and correctness) rather than CDI.

Note on computation: CDI standard deviations (C7: 0.261; C10b: 0.086; C11: 0.163) and means are taken directly from the scale study output file (`outputs/protocol_test/protocol_test_20260512_073353.json`, $n = 321$ conversations). Welch t-statistics and Cohen's d values are computed analytically from these reported statistics rather than from individual-level data; exact p-values assume approximate normality of CDI distributions within each condition.

---

5. Discussion

5.1 Epistemic Interdependence as a Design Principle

The central finding across all pilot iterations is that CDI is a function of split quality, not instruction quality. No amount of collaborative framing can compensate for a data split: when the information distribution allows either agent to formulate a complete solution, CDI remains low regardless of whether agents are instructed to collaborate deeply.

This finding has direct implications for activity design in mathematics education. The behavioral solo-verification procedure (§3.5) operationalizes the epistemic fidelity criterion at the task level: a teacher designing a jigsaw activity should ask not "have I given each student different information?" but "can either student solve this without the other's contribution?" The 50% failure rate in our generated corpus — where half of algorithmically produced splits turned out to be behaviorally trivial — demonstrates that this question cannot be answered by inspection alone. Automated behavioral verification, as implemented here, is a practical requirement for any collaborative task design pipeline using LLM agents.

The moderator finding extends this principle: even among genuinely INTERDEPENDENT splits, the protocol that maximises collaborative quality (C10b) reduces task accuracy for SPLIT-F problems (functional chains in probability and counting). This suggests a second-order epistemic fidelity criterion at the protocol level: the coordination structure must be matched to the epistemic demands of the split, not applied uniformly. A plan-first protocol is well-matched to multi-step reasoning problems (SPLIT-C, SPLIT-A) but poorly matched to single-step combination problems (SPLIT-F) where the optimal path is direct information exchange followed by a single computation.

5.2 Task-Role Chain Design and the Epistemic Fidelity Problem

The four-iteration development history reveals that epistemic validity in LLM-based CPS simulation requires solving two distinct problems: (1) ensuring the system prompt does not prescribe the

collaboration it is meant to measure (minimal framing principle), and (2) ensuring the information structure actually prevents unilateral solution (task-role chain design, not knowledge restriction). The v3 experience shows that minimal framing alone is insufficient: if the goal-anchor carries the full problem text, agents solve it immediately, rendering the split irrelevant regardless of how epistemically sound the split design is. The v4 combination — minimal system prompt + task-role chain packets + shared_context goal-anchor — closes both gaps simultaneously. For LLM agents with unrestricted mathematical knowledge, task-chain interdependence (functional: agent A's output is agent B's required input) is more robust than knowledge-restriction interdependence (epistemic: agent A does not know what B knows), because task-chain cannot be bypassed by an agent who already possesses the full solution procedure.

The task-role chain design principle (§2.6) extends beyond this study. In any collaborative task where participants have unrestricted access to background knowledge — as is the case with LLM agents, and arguably with expert human collaborators as well — knowledge restriction as a design strategy is fragile. An LLM that "knows" the answer but is instructed to act as if it doesn't will produce qualitatively different collaborative behaviour than one whose information packet genuinely requires the partner's output to proceed. The task-chain approach sidesteps this fragility by making the dependency functional rather than epistemic: the agent cannot compute its assigned output without a specific numerical input from the partner, and this constraint is structural rather than instructional. The design lesson for human-learner settings is analogous: assign computational roles rather than information silos, and the collaborative dependency will be preserved even as students develop mathematical fluency that might otherwise bypass the split.

5.3 CY and the Process-Outcome Relationship

The observed independence of CDI and correctness (r=-0.015 in pilot v2) has two interpretations. First, it may indicate that the conditions do not yet produce epistemic splits of sufficient quality for collaboration to be necessary for correct outcomes. Second, it may reflect genuine productive failure dynamics: agents that collaborate deeply on difficult problems engage in a form of mathematical struggle that is educationally valuable even without immediate success (Kapur, 2012).

The scale study resolves this ambiguity partially. Under C10b, the independence of CDI and correctness is not an artifact of poor split quality: the 36 problems used in the scale study were all behaviorally INTERDEPENDENT (verified by solo-verification), and C10b's near-ceiling CDI (0.967 $\pm$ 0.086) indicates genuine collaborative engagement. The residual correctness ceiling (44%) reflects the difficulty of the MATH benchmark problems at levels 3–5, not a failure of the collaborative process. This interpretation is consistent with the productive failure model: C10b conversations that reach CDI=1.0 but produce an incorrect answer are exhibiting the target state of deep collaborative engagement — the collaboration is working, but the problem is hard. From a learning-outcomes perspective, these are precisely the conversations most likely to produce durable conceptual learning (Kapur, 2016).

The CY metric formalises this interpretation by rewarding high-CDI conversations regardless of correctness (though less than COUPLING outcomes), and applying a penalty only to low-CDI correct responses (TRIVIAL quadrant) that indicate collaboration was not necessary for the outcome. In a learning context where the goal is collaborative skill development, a CY-ranked activity design would select for problems that reliably produce COUPLING and PRODUCTIVE_FAILURE outcomes, avoiding both TRIVIAL (too easy) and COLLAPSE (too hard or poor split).

5.4 Implications for Collaborative Activity Design

The framework suggests a design workflow for practitioners: (1) analyze the problem for natural *computational chains* — sequences where one sub-result is required to formulate the next sub-problem (M1 bottleneck analysis); (2) assign each participant a role in the chain, specifying inputs, the computation to perform, outputs to share, and what is needed from the partner; (3) ensure the shared context contains only the collaborative goal, not the problem data (goal-anchor discipline); (4) apply minimal framing — state the collaborative context and exchange permission, nothing more; (5) close with a structural Phase D verification that requires independent answer declaration from each participant. Step 2 operationalizes the task-role chain principle: activity designers should ask "what does each participant need to produce for the next step?" rather than "what information should each participant keep private?"

A natural extension of this design workflow is grounding the coordination protocol in Socially Shared Regulation (SSR) theory. The C12 condition (not included in the scale study) extends C11 with three SSR-derived rules: R1 (joint state check at phase transitions) operationalizes SSR-Monitoring, R2 (grounded evaluation with epistemic justification) operationalizes SSR-Evaluation, and R3 (initiative transfer after three or more consecutive proposals by one agent) operationalizes SSR-Participation regulation — the three core SSR functions identified by Järvelä and Hadwin (2013). The correspondence is direct: SSR-Monitoring requires that both participants share a representation of the current collaborative state before moving to the next phase; SSR-Evaluation requires that assessments of intermediate outputs be grounded in the problem structure rather than social agreement; and SSR-Participation regulation ensures that epistemic contributions remain distributed across the dyad rather than converging on a single agent. However, C12 exhibited high variance in the pilot that precluded its inclusion in the scale study. This finding is itself theoretically informative: if SSR scaffolding is highly problem-dependent in its effects, this suggests that the three SSR functions are not uniformly triggered by the same problem features, and that the conditions under which each function becomes salient deserve systematic investigation. Future work should examine whether the variance in C12 correlates with split pattern (e.g., SPLIT-F functional chains may require more SSR-Monitoring than SPLIT-B boundary problems) and whether targeted SSR scaffolding can be applied selectively rather than uniformly across the protocol.

5.5 Limitations

- **LLM agents as learner proxies**: LLM agents have broad mathematical knowledge not constrained to any developmental level. The study demonstrates the framework's measurement properties with LLM agents; validity with human learners requires separate investigation.
- **Pilot scale**: n=4 problems $\times$ 5 conditions = 20 conversations per iteration. Statistical power is limited; conclusions are primarily methodological.
- **Domain specificity**: Problems are competition-style algebra/trigonometry. Generalizability to other mathematical domains requires testing.
- **LLM agents as learner proxies**: Expanded note. LLM agents have unrestricted mathematical knowledge not limited to any curriculum level, and their response to collaborative framing may differ systematically from human students. Task-role chain interdependence is functional for LLMs (the agent *cannot* produce the answer without a specific numerical input), but human students may experience the interdependence differently — as a source of frustration, motivation, or scaffolded discovery depending on their ZPD (Vygotsky, 1978). The

validity of CDI as a predictor of human collaborative quality is an open empirical question that requires a study with real students at a defined curriculum level.

---

6. Conclusions and Future Work

This paper has introduced the CPP framework, the CDI metric, the CY composite, and the minimal framing principle as theoretical and methodological contributions to the design and study of AI-generated collaborative mathematical tasks. The scale study (n=321 conversations, 36 INTERDEPENDENT problems, 3 conditions) demonstrates that coordination protocol design produces large and reliable improvements in collaborative engagement: the plan-first student-simulation protocol (C10b) raises mean CDI from 0.747 to 0.967 relative to the student-simulation baseline (C7), with variance compressed from ± 0.261 to ± 0.086. Eighty percent of C10b conversations activate all 12 PISA CPS cells, compared to 28% under C7. The framing-effects spectrum (CEXP = 0.294 through C10b = 0.967) establishes that expert framing suppresses collaborative engagement while the combination of student-simulation framing and plan-first structure is the most effective protocol tested. These results confirm that collaborative behavior in LLM-agent dyads is a function of both task structure (epistemic split quality) and coordination protocol, and that both components can be systematically designed.

Future work includes: (1) validation with human learners at defined curriculum levels, applying domain restriction rather than persona to the agent when used as a peer tutor; (2) extension to n=3 and n=4 agent groups to test phase dynamics at higher n; (3) automated epistemic fidelity scoring from M5 output to scale solo-verification beyond the current manual procedure; (4) targeted SSR scaffolding by split pattern, following the C12 high-variance finding; (5) curriculum mapping to identify epistemic split opportunities across Calculus 1 topic areas beyond the algebra and trigonometry problems studied here.

References

Aronson, E., Blaney, N., Stephan, C., Sikes, J., & Snapp, M. (1978). *The jigsaw classroom.* Sage Publications.

Chi, M. T. H., & Wylie, R. (2014). The ICAP framework: Linking cognitive engagement to active learning outcomes. *Educational Psychologist*, 49(4), 219–243.

Dillenbourg, P. (1999). What do you mean by collaborative learning? In P. Dillenbourg (Ed.), *Collaborative Learning: Cognitive and Computational Approaches* (pp. 1–19). Pergamon/Elsevier.

Greiff, S., Wüstenberg, S., Csapó, B., Demetriou, A., Hautamäki, J., Graesser, A. C., & Martin, R. (2014). Domain-general problem solving skills and education in the 21st century. *Educational Research Review*, 13, 74–83.

Järvelä, S., & Hadwin, A. F. (2013). New frontiers: Regulating learning in CSCL. *Educational Psychologist*, 48(1), 25–39.

Johnson, D. W., & Johnson, R. T. (1989). *Cooperation and competition: Theory and research.* Interaction Book Company.

Kapur, M. (2016). Examining productive failure, productive success, unproductive failure, and unproductive success in learning. *Educational Psychologist*, 51(2), 289–299.

Kirschner, F., Paas, F., & Kirschner, P. A. (2011). Task complexity as a driver for collaborative learning efficiency: The collective working-memory effect. *Applied Cognitive Psychology*, 25(4), 615–624.

OECD (2015). *PISA 2015 Assessment and Analytical Framework: Science, Reading, Mathematics and Collaborative Problem Solving.* OECD Publishing. https://doi.org/10.1787/9789264281820-en

OECD (2017). *PISA 2015 Results (Volume V): Collaborative Problem Solving.* OECD Publishing.

Roschelle, J., & Teasley, S. D. (1995). The construction of shared knowledge in collaborative problem solving. In C. O'Malley (Ed.), *Computer Supported Collaborative Learning* (pp. 69–97). Springer.

Sfard, A. (1998). On two metaphors for learning and the dangers of choosing just one. *Educational Researcher*, 27(2), 4–13.

Szewkis, E., Nussbaum, M., Rosen, T., Abalos, J., Denardin, F., Caballero, D., Tagle, A., & Alcoholado, C. (2011). Collaboration within large groups in the classroom. *International Journal of Computer-Supported Collaborative Learning*, 6(4), 561–575.

Vygotsky, L. S. (1978). *Mind in society: The development of higher psychological processes.* Harvard University Press.

Barron, B. (2003). When smart groups fail. *Journal of the Learning Sciences*, 12(3), 307–359.

Bereiter, C., & Scardamalia, M. (2010). Can children really create knowledge? *Canadian Journal of Learning and Technology*, 36(1).

Chi, M. T. H., Siler, S. A., Jeong, H., Yamauchi, T., & Hausmann, R. G. (2001). Learning from human tutoring. *Cognitive Science*, 25(4), 471–533.

Csikszentmihalyi, M. (1990). *Flow: The psychology of optimal experience.* Harper & Row.

Graesser, A. C., Cai, Z., Baer, W., Olney, A., & Hu, X. (2018). Collaborative problem solving in a conversational agent. In *International Handbook of the Learning Sciences* (pp. 96–105). Routledge.

Greiff, S., Wüstenberg, S., & Funke, J. (2012). Dynamic problem solving: A new assessment perspective. *Applied Psychological Measurement*, 36(3), 189–213.

Hmelo-Silver, C. E. (2004). Problem-based learning: What and how do students learn? *Educational Psychology Review*, 16(3), 235–266.

Hong, S., Zheng, X., Chen, J., et al. (2023). MetaGPT: Meta programming for a multi-agent collaborative framework. *arXiv:2308.00352.*

Järvelä, S., & Hadwin, A. F. (2013). New frontiers: Regulating learning in CSCL. *Educational Psychologist*, 48(1), 25–39.

Kapur, M. (2012). Productive failure in learning the concept of variance. *Instructional Science*, 40(4), 651–672.

Kirschner, P. A., Sweller, J., & Clark, R. E. (2006). Why minimal guidance during instruction does not work. *Educational Psychologist*, 41(2), 75–86.

Koedinger, K. R., & Corbett, A. T. (2006). Cognitive tutors: Technology bringing learning sciences to the classroom. In R. K. Sawyer (Ed.), *The Cambridge Handbook of the Learning Sciences* (pp. 61–77). Cambridge University Press.

Liang, T., He, Z., Jiao, W., et al. (2023). Encouraging divergent thinking in large language models through multi-agent debate. *arXiv:2305.19118*.

Mehan, H. (1979). *Learning lessons: Social organization in the classroom*. Harvard University Press.

Mercer, N. (1996). The quality of talk in children's collaborative activity in the classroom. *Learning and Instruction*, 6(4), 359–377.

OECD (2015). *PISA 2015 Assessment and Analytical Framework*. OECD Publishing.

Park, J. S., O'Brien, J. C., Cai, C. J., Morris, M. R., Liang, P., & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. In *Proceedings of UIST 2023* (pp. 1–22). ACM.

Roschelle, J. (1992). Learning by collaborating: Convergent conceptual change. *Journal of the Learning Sciences*, 2(3), 235–276.

Sfard, A. (1998). On two metaphors for learning and the dangers of choosing just one. *Educational Researcher*, 27(2), 4–13.

Scardamalia, M., & Bereiter, C. (2014). Knowledge building and knowledge creation: Theory, pedagogy, and technology. In R. K. Sawyer (Ed.), *The Cambridge Handbook of the Learning Sciences* (2nd ed., pp. 397–417). Cambridge University Press.

Stahl, G. (2006). *Group cognition: Computer support for building collaborative knowledge*. MIT Press.

Suthers, D. D. (2006). Technology affordances for intersubjective meaning making. *International Journal of Computer-Supported Collaborative Learning*, 1(3), 315–337.

Volet, S., Vauras, M., & Salonen, P. (2009). Self- and social regulation in learning contexts. *Educational Psychologist*, 44(4), 215–226.

Wu, Q., Bansal, G., Zhang, J., Wu, Y., Zhang, S., Zhu, E., Li, B., Jiang, L., Zhang, X., & Wang, C. (2023). AutoGen: Enabling next-gen LLM applications via multi-agent conversation. *arXiv:2308.08155*.

Wegerif, R. (2007). *Dialogic education and technology: Expanding the space of learning*. Springer.

Windschitl, M. (2002). Framing constructivism in practice as the negotiation of dilemmas. *Review of Educational Research*, 72(2), 131–175.

Zimmerman, B. J. (2000). Self-efficacy: An essential motive to learn. *Contemporary Educational Psychology*, 25(1), 82–91.

Webb, N. M. (1991). Task-related verbal interaction and mathematics learning in small groups. *Journal for Research in Mathematics Education*, 22(5), 366–389.